PhD. Abdelkrim Imghoure

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Status: Permanent Resident of Canada | French Citizen

System Safety Expert

9 years of experience in functional safety and cybersecurity in the automotive industry, complemented by PhD research focused on automotive cybersecurity.

PROFESSIONAL EXPERIENCE

CS Group, Montreal, Canada
System Safety Expert - Full time – Division: Autonomous systems Division 05/2024 – Present

- Lead functional safety and cybersecurity analysis activities in compliance with safety and cybersecurity standards.
- Create a process for CSMS, TARA, cybersecurity concept and attack tree analysis.
- Provide training materials in Functional Safety and Cybersecurity according to ISO 26262, ISO 21448 and ISO 21434.
- Carried out safety and cybersecurity activities for the following projects:
  - **Torc Robotics - USA**: Created Hazard Identification and Risk Evaluation analysis, using HAZAOP, HARA, STPA analysis for self-driving truck (perception, planning, control) according to ISO 21448 (part 6).
  - **Hexagon Purus - Canada**: Conduct I3 confirmation reviews for Item Definition – HARA – System architecture, FSC, FTA and TSC for several safety-related systems for electric vehicle according ISO 26262.
  - **Harbinger - USA**: Created Item Definition, HAZOP, HARA, FTA and FSC for Telematic control unit with FOTA function to perform updates on the operational vehicle domains: Chassis, Steering, Powertrain, Infotainment, ADAS for electric vehicle.
  - **BRP Canada**: Created and reviewed TARA, Attack Tree Analysis for Rechargeable Energy Storage System (RESS), Vehicle Stability System (VSS), Start-stop system and Lighting system.
- Perform Safety analysis: SW FTA, SW FMEA – SW Safety analysis: DFA using QNX OS and Hypervisor.

Standards: ISO 26262, ISO 21448, ISO 21434, UL4600 and ISO 8800.

Honda Development and Manufacturing of America, Marysville, Ohio
Contractor for ALTEN Technology USA - Full time
Functional Safety Engineer – Division: Advanced: Safety Controls Division 01/2023 – 03/2024

- Analyzed new requirements specification and potential changes to identity safety and cybersecurity impacts on ISO 26262 and ISO 21434 work products for active safety systems (ASIL D).
- Analyzed and reviewed OEM and Supplier documents to ensure compliance with ISO 26262, ISO 21448, ISO 21434, AUTOSAR and ASPICE.
- Conducted meetings with testing teams to prepare testing and validation documents.
- Conducted verification and confirmation reviews, and functional safety milestone assessments with Japan.

Continental USA, Troy, Michigan
Contractor for ALTEN Technology USA – Full time
Functional Safety Engineer – Division: Vehicle Networking and Information 03/2019 – 12/2022

- Led functional safety and cybersecurity analysis activities for Hardware and Software in compliance with ISO 26262, ISO 21448, ISO 21434, IEEE 1609.2 related to V2X technology, AUTOSAR and ASPICE on FCA, Ford, GM projects (ASIL A – ASIL B – ASIL D)
- Created Safety Plan - Hazard Analysis and Risk Assessment (HARA) - Functional Safety Concept (FSC) and Technical Safety Concept (TSC) -update and maintain FTA, DFMEA and FMEDA analysis to evaluate HW architectural metrics – review and verify SW safety concept.
- Perform Safety analysis: SW FTA, SW FMEA – SW Safety analysis: DFA using QNX OS.

- Reviewed Threat Analysis and Risk Assessment (TARA) and Cybersecurity concept.
- Conducted verification and confirmation reviews, and functional safety milestone assessments with FCA and GM.

Renault Nissan Mitsubishi, Guyancourt, France
Contractor for Ortec Group - Full time
Quality and Safety Engineer – Department of Engineering & Advanced Systems 11/2016 – 01/2019

- Led functional safety analysis activities in compliance with ISO 26262 and ISO 21448.
- Carried out quality assurance activities on Renault/Nissan projects.
- Created statistical models that predict failure rates based on feedback.
- Created computer programs to track the Unique List of Problems (LUP) until resolution.
- Conducted verification and confirmation reviews, and functional safety milestone assessments with Renault.

PROFESSIONAL CERTIFICATES

Controlled Goods Program – Exemption Certificate 05/2024
Issued by CS GROUP - Canada Inc.
Authorized to access, possess, and transfer controlled goods, including U.S.-origin military vehicle technology (2-22.a) and military ground vehicles (2-6.a, 2-6.b).

IBM AI Engineering Professional Certificate (*4 months training*) 12/2024 – 04/2025
IBM AI Engineering Professional Certificate (V3) - Credly

Cybersecurity Specialist (TÜV Rheinland) # 912 / 23 (*4 hours exam)* 2023
https://www.certipedia.com/search/matching_cysec_specialist_certificates?&locale=en&q=imghoure

Functional Safety Engineer (TÜV Rheinland) # 28235 / 24 (*4 hours exam)* 2023
https://www.certipedia.com/search/matching_fs_engineer_certificates?&locale=en&q=imghoure

NCEES Electrical and Computer: Electronics, Controls, and Communications (PE exam) (*8 hours exam)* 2023
https://account.ncees.org/rn/2263506-1595401-da2be45

NCEES Fundamentals of Engineering (FE exam) – (*6 hours exam)* 2023
https://account.ncees.org/rn/2263506-1548184-a4570c9

Agile Boot Camp: ICP Fundamentals Certificate 2021

AWARD AND RESEARCH

American Journal of Networks and Communications: Science Publishing Group 06/2024 – Present
American Journal of Networks and Communications
Peer-reviewer

Certificate of reviewing award, Information Fusion – Elsevier 06/2024 – 12/ 2024
Description: *Recognition for contributions as a reviewer in Cybersecurity*

Certificate of reviewing award, Vehicular communications – Elseiver 09/2022 – 07/2023
Description: *Recognition for contributions as a reviewer in Automotive Cybersecurity*

EDUCATION

Mohammed V University in Rabat, Rabat, Morocco 2024
Ph.D in Computer Science
Specialty: *Cryptography and Information Security*
Thesis title: *New Authentication Processes in Vehicular Ad-Hoc Network*

Ecole Supérieure des Sciences et Technologies de l'Ingénieur de Nancy (ESSTIN), Nancy, France
(Graduate School of Engineering Sciences & Technologies)
Master's Degree in Engineering Science 2015

Université de Strasbourg, Strasbourg, France
Bachelor's Degree in Mathematics 2012

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PUBLICATIONS IN Q1 JOURNALS

IMGHOURE, Abdelkrim, EL-YAHYAOUI, et Ahmed OMARY, Fouzia. Strong non-repudiation certificateless scheme with an aggregate signature in vehicular ad-hoc network, Franklin Open (2024)
https://www.sciencedirect.com/science/article/pii/S2773186324001361#bib0003

IMGHOURE, Abdelkrim, OMARY, Fouzia, et EL-YAHYAOUI, Ahmed. Hybrid Cryptography-based Scheme with Conditional Privacy-Preserving Authentication and Memory-based DOS Resilience in V2X. Vehicular Communications, 2024, p. 100810.
https://www.sciencedirect.com/science/article/abs/pii/S2214209624000858

IMGHOURE, Abdelkrim, OMARY, Fouzia, et EL-YAHYAOUI, Ahmed. Schnorr-based conditional privacy-preserving authentication scheme with multisignature and batch verification in vanet. Internet of Things, 2023, vol. 23, p. 100850.
https://www.sciencedirect.com/science/article/abs/pii/S2542660523001737

IMGHOURE, Abdelkrim, EL-YAHYAOUI, Ahmed, et OMARY, Fouzia. ECDSA-based certificateless conditional privacy-preserving authentication scheme in Vehicular Ad Hoc Network. Vehicular Communications, 2022, vol. 37, p. 100504.
https://www.sciencedirect.com/science/article/abs/pii/S2214209622000511

PROFESSIONAL MEMBERSHIPS

- Senior Member, IEEE
- Member, SAE

PROFICIENCES

Analysis tools

Medini - DOORS- JIRA Board – IMS Configuration Management – Isograph Fault Tree+ - Medini Analyze – Confluence
CI/CD Tool Chain: Jenkins, Git, GitLab

Languages

English, French

Computer Programs

Penetration testing tools: Python - Kali - Nmap - Metasploit - Hydra- Burp Suite – Wireshark – Gobuster – Mimikatz - John the Ripper.
Programming languages: C Language - C++ Language – Python - VBA - JAVA - R (Statistic) – MATLAB.
Databases: MySQL, Microsoft Access

AI libraries: Keras – PyTorch – Tensorflow